Your Prompt Is Your Result: How Pipeline Choices Manufacture Metacognitive Evaluation Findings

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Abstract

We report five empirical findings that emerged from diagnosing our own prior evaluation pipeline for LLM metacognition on biomedical claim verification (PubMedQA). The two most consequential findings are: (1) truncating the generation limit to `max_tokens=80` artificially suppressed Llama-3.3-70b accuracy by 38 percentage points ($0.247 \rightarrow 0.631$), and (2) supplying an explicit action-mapping table in the prompt drives Cohen's κ between predicted and rule-derived actions to 1.000 across 600 responses from two models, eliminating any measurable variation in the control stage. Three further findings concern evidence length and sample composition driving 88% of the abstention artifact, a stable calibration gap of $+0.25$ that persists across all pipeline variants for GPT-4.1-mini, and a systematic failure to detect genuine ambiguity (INCONCLUSIVE recall 12–25% against a gold rate of 33%). Removing the threshold table produces a significant shift in action distributions for both models (Stuart–Maxwell $p < 0.01$), revealing genuine behavioral differences masked by the original design. We derive a six-field reporting checklist for metacognitive evaluation pipelines. All data and code are available at https://anonymous.4open.science/r/pipeline-artifacts-8520/.

1 Introduction

Evaluating whether large language models (LLMs) can monitor and regulate their own uncertainty has become an active research area (Kadavath et al., 2022; Xiong et al., 2024; Tian et al., 2023). A common paradigm asks a model to (1) classify a claim, (2) estimate its own error probability, and (3) choose a control action (e.g., commit, abstain, or seek evidence) based on that estimate. The implicit assumption is that the third stage measures something about the model's internal policy.

We show that this assumption can fail silently due to ordinary pipeline choices. The findings reported here emerged from diagnosing our own prior evaluation pipeline—not from auditing others' work. We describe the pipeline, identify the confounds, isolate each effect through controlled ablations, and propose a reporting checklist to prevent recurrence.

Our contributions are:

1. A documented case in which `max_tokens=80` produced a 38-point accuracy artifact in a 70B-parameter model (§4.1). (A test run with Gemini-2.5-Flash yielded 998/1000 parse failures, further confirming that measured performance often reflects parser compliance rather than model capability).

2. Empirical proof that supplying an explicit threshold table in the prompt renders the control stage a deterministic function of the monitoring stage ($\kappa = 1.000$), eliminating measurable variation (§4.3).

3. Evidence that evidence length and sample composition, not the threshold table, drive 88% of the abstention shift; threshold-table removal accounts for only $+3.7$ pp (§4.2).

4. A stable calibration gap ($+0.25$) that is robust to all pipeline variants tested (§4.4).

5. A systematic under-detection of genuine ambiguity (INCONCLUSIVE recall 12–25%) (§4.5).

6. A six-field reporting checklist for metacognitive evaluation pipelines (§5).

2 Background and Related Work

LLM metacognition. Kadavath et al. (2022) showed that LLMs can estimate the probability that their own answers are correct, and that these estimates are moderately calibrated. Xiong et al. (2024) extended this to chain-of-thought settings, finding that verbalized confidence is often overconfident. Tian et al. (2023) demonstrated that elicitation format strongly affects expressed confidence. Griot et al. (2025) found that LLMs lack essential metacognition for reliable medical reasoning, a finding our results partially corroborate and partially complicate.

Calibration in NLP. Guo et al. (2017) established that neural networks are systematically overconfident and that temperature scaling partially corrects this. Desai and Durrett (2020) showed that pre-trained language models inherit miscalibration from fine-tuning. Wang (2026) found that supplying calibration scores to models during inference does not reliably improve downstream decision quality—a result whose mechanism our findings help explain: if the prompt already encodes the mapping from score to action, the score carries no additional information.

Prompt sensitivity. Sclar et al. (2024) showed that minor formatting changes in prompts cause large swings in benchmark performance. Mizrahi et al. (2024) demonstrated that multi-prompt evaluation is necessary for reliable assessment. Lu et al. (2021) documented order sensitivity in few-shot prompting. These findings motivate our focus on prompt-level confounds.

Negative results and evaluation validity. Bender et al. (2021) raised concerns about evaluation validity in large-scale NLP. Zhang et al. (2026) introduced prefix branching to measure behavioral responsiveness to uncertainty signals from identical trajectory states. Our work differs in that we report a failure mode of standard elicitation pipelines rather than a positive intervention result. Wang (2026) proposed a hierarchical benchmark for metacognitive calibration; our checklist (§5) is intended to complement such benchmarks by specifying what pipeline parameters must be reported.

3 Task and Pipeline

3.1 Task: Biomedical Claim Verification with Metacognitive Control

We evaluate LLMs on PubMedQA (Jin et al., 2019), a biomedical question-answering dataset with three gold labels: SUPPORTED, REFUTED, and INCONCLUSIVE. Each model instance receives a claim and PubMed evidence and must complete three tasks in a single prompt:

- **Task 1 (Decision):** Classify the claim as SUPPORTED, REFUTED, or INCONCLUSIVE.
- **Task 2 (Monitoring):** Estimate the probability that the Task 1 decision is wrong (a scalar in [0, 1]).
- **Task 3 (Control):** Choose an action from {COMMIT, ABSTAIN, SEEK_EVIDENCE, REVISE}.

The control rate (CR) is the fraction of responses in which the model chooses a safety-oriented action (ABSTAIN or SEEK_EVIDENCE). The calibration gap is defined as actual error rate $-\ \hat{p}_{\text{error}}$, where $\hat{p}_{\text{error}}$ is the mean declared error probability; a positive gap indicates overconfidence. The rule-agreement κ measures Cohen's κ between model-chosen actions and rule-derived actions (the action implied by applying the threshold table to the declared error probability).

3.2 Models

We evaluate two models: GPT-4.1-mini (OpenAI, accessed July 2026) and Llama-3.3-70b (Meta, served via Together AI, accessed July 2026). Temperature is set to 0 for all experiments.

3.3 Four Protocol Conditions

To isolate the confounds in our original pipeline, we decompose it into four conditions.

Protocol A0 (Original Pipeline). Used `max_tokens=80`, evidence truncated to ≈800 characters, and a natural sample of 1000 PubMedQA cases (gold distribution: 55.2% Supported, 33.8% Refuted, 11.0% Inconclusive). Task 3 included an explicit threshold table:

```
COMMIT: error_prob < 0.30
ABSTAIN: 0.30 <= error_prob < 0.60
SEEK_EVIDENCE: error_prob >= 0.60
```

Protocol A1 (Isolating `max_tokens`). Identical to A0, but with `max_tokens=300`. This isolates the effect of the generation limit.

Protocol A2 (Isolating Evidence and Sample). Uses `max_tokens=300`, full evidence (up to 2400 characters), and a stratified sample (100 cases per gold label) with a fixed random seed. The threshold table is retained.

Protocol B (Isolating the Threshold Table). Identical to A2 (same 300 stratified cases, full evidence), but removes the threshold table from Task 3, replacing it with:

```
Choose the most appropriate action.
No numerical thresholds are provided.
```

The A0 $\to$ A1 comparison isolates the token limit. The A1 $\to$ A2 comparison shows the joint effect of evidence length and sample stratification. The A2 $\to$ B comparison isolates the causal effect of the threshold table.

4 Results

4.1 Finding 1: `max_tokens=80` Suppressed Llama Accuracy by 38 Points

Table 1 reports accuracy, INCONCLUSIVE rate, and action distribution across protocols. Llama-3.3-70b accuracy rises from 0.247 (Protocol A0) to 0.631 (Protocol A1), a difference of 38.4 percentage points. The mechanism is straightforward: at `max_tokens=80`, Llama frequently begins responses with a reasoning preamble before stating DECISION:, causing truncation before the parseable content. The parser then returns the default label INCONCLUSIVE, inflating that class to 78.9% in A0. This is a silent default assignment, not a measured parse failure (the parser successfully returned a result, but the result was an artifact of truncation).

GPT-4.1-mini shows a smaller but consistent improvement (0.462 $\to$ 0.512), suggesting it more reliably front-loads the structured output. In A1, GPT-4.1-mini (0.512) remains below the majority-class baseline (0.552) while Llama-3.3-70b (0.631) exceeds it.

4.2 Finding 2: Evidence Length and Sample Composition Drive 88% of the Abstention Artifact

Figure 1 shows the action distribution across protocols and models. Comparing Protocol A1 (truncated evidence, $N = 1000$) to Protocol B (full evidence, no threshold, $N = 300$) for GPT-4.1-mini:

- ABSTAIN rate: 34.7% $\to$ 9.7% (-25.0 pp)
- COMMIT rate: 65.1% $\to$ 88.3% ($+23.2$ pp)

To decompose this shift, we compare Protocol A1 to Protocol A2 (full evidence, with threshold table retained):

- ABSTAIN rate: 34.7% $\to$ 6.0% (-28.7 pp)

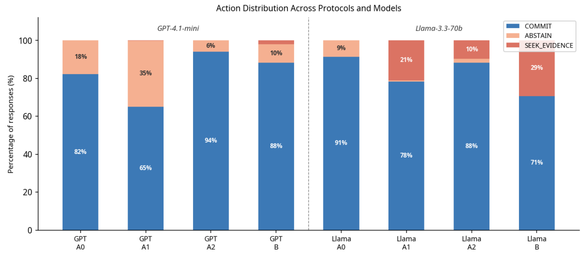

Figure 1: Action distribution across protocols and models. With the threshold table present (A1–A2), action distributions are deterministic functions of declared error probability. The shift from A2 to B (threshold table removed) reveals genuine behavioral differences: GPT-4.1-mini shifts from 94% COMMIT / 6% ABSTAIN (A2) to 88% COMMIT / 10% ABSTAIN (B); Llama-3.3-70b shifts from 88% COMMIT / 2% ABSTAIN / 10% SEEK (A2) to 71% COMMIT / 0% ABSTAIN / 29% SEEK (B). Evidence length and sample composition jointly drive 88% of the total abstention shift.

And Protocol A2 to Protocol B (threshold table removed):

- ABSTAIN rate: 6.0% $\to$ 9.7% ($+3.7$ pp).

The full abstention-rate drop from A1 to B is therefore driven predominantly by the evidence-length and sample-composition change (-28.7 pp), which explains 88% $(28.7/(28.7+3.7))$ of the shift. We cannot separate these two factors, since A1 and A2 differ in both. The threshold table removal actually slightly increases abstention ($+3.7$ pp). The action distribution that our original pipeline attributed to model behavior was predominantly a consequence of evidence truncation, mediated by the supplied threshold rule: shorter evidence raised declared error probabilities, and the rule mechanically converted those into ABSTAIN.

4.3 Finding 3: The Threshold Table Eliminates Measurable Control Variation

Table 2 reports rule-agreement and Cohen's κ between model-chosen actions and rule-derived actions. With the threshold table present ($\kappa = 1.000$ in A2 for both models; $\kappa = 0.998$ for GPT in A1), there is effectively zero variance in the control stage: every model action is a deterministic function of the declared error probability. This means that CR and any delta computed over control actions are not measurements of model behavior—they are algebraic consequences of the monitoring stage output passed through the supplied mapping.

Without the threshold table, the models diverge substantially. GPT-4.1-mini remains largely con-

Protocol	Model	N	Acc.	INC%	Baseline	COMMIT%	ABSTAIN%	SEEK%
A0 (orig.)	GPT-4.1-mini	1000	0.462	47.7%	0.552	82.2%	17.8%	0.0%
	Llama-3.3-70b	1000	0.247	78.9%	0.552	91.4%	8.6%	0.0%
A1 (tokens)	GPT-4.1-mini	1000	0.512	42.1%	0.552	65.1%	34.7%	0.2%
	Llama-3.3-70b	1000	0.631	21.2%	0.552	78.2%	0.7%	21.1%
A2 (evid.)	GPT-4.1-mini	300	0.590	15.0%	0.333	94.0%	6.0%	0.0%
	Llama-3.3-70b	300	0.567	10.7%	0.333	88.3%	2.0%	9.7%
B (no thr.)	GPT-4.1-mini	300	0.600	14.3%	0.333	88.3%	9.7%	2.0%
	Llama-3.3-70b	300	0.577	10.0%	0.333	70.7%	0.0%	29.3%

Table 1: Accuracy, INCONCLUSIVE prediction rate, majority-class baseline, and action distribution across protocols. A0/A1 use a natural sample ($N = 1000$); A2/B use a stratified sample ($N = 300$). The parser returned a value for every response; however, under A0 truncation it returned the INCONCLUSIVE default rather than a parsed decision. We report this as a silent default, not a parse success. Under A0, the ACTION field appears after DECISION in the output format; where truncation prevented a parsed decision, the action is likewise a parser default. A0 action percentages should be read as pipeline output, not model behavior.

Model	Protocol	Agree	κ	95% CI
GPT	A1	0.999	0.998	–
	A2	1.000	1.000	–
	B (no thr.)	0.970	0.860	[0.750, 0.940]
Llama	A1	1.000	1.000	–
	A2	1.000	1.000	–
	B (no thr.)	0.800	0.406	[0.300, 0.511]

Table 2: Rule-agreement between model actions and threshold-derived rule actions. $\kappa = 1.000$ in A2 is exact (perfectly diagonal, zero off-diagonal cases for both models, $n = 600$). GPT shows a single deviation in A1 ($\kappa = 0.998$). Bootstrap CIs ($N = 10000$) are provided for Protocol B. A Stuart–Maxwell test of marginal homogeneity between A2 and B action distributions is significant for both models ($p < 0.01$; note: Llama uses ABSTAIN zero times in B, producing a zero marginal for that row).

sistent ($\kappa = 0.860$), while Llama-3.3-70b departs significantly ($\kappa = 0.406$), revealing a genuine behavioral difference that the threshold table had masked. A Stuart–Maxwell test of marginal homogeneity between the A2 and B action distributions confirms the shift is statistically significant for both GPT ($\chi^2 = 79.8, p < 0.001$) and Llama ($\chi^2 = 10.6, p = 0.005$).

Table 3 shows the confusion matrix for Llama in Protocol B. Of the 270 cases in which the rule would prescribe COMMIT, Llama selects SEEK_EVIDENCE in 58; it selects ABSTAIN zero times. This reveals a genuine behavioral preference: Llama treats "I may be wrong" and "this evidence is insufficient" as orthogonal, whereas the threshold table orders them by defining SEEK_EVIDENCE as $p \geq 0.60$.

Rule-derived action	COMMIT	SEEK	Total
COMMIT	212	58	270
ABSTAIN	0	2	2
SEEK_EVIDENCE	0	28	28
Total	212	88	300

Table 3: Confusion matrix for Llama-3.3-70b in Protocol B (no threshold table). Rows = rule-derived action; columns = model-chosen action. Of the 270 cases where the rule prescribes COMMIT (declared error probability below 0.30), Llama selects SEEK_EVIDENCE in 58—requesting additional evidence while judging itself unlikely to be wrong. It selects ABSTAIN zero times.

4.4 Finding 4: Calibration Gap Is Stable Across All Pipeline Variants

Table 4 and Figure 2 report the calibration gap (actual error rate minus declared error probability) across protocols. A positive gap indicates overconfidence (the model underestimates its own error rate).

GPT-4.1-mini's calibration gap is remarkably stable at approximately $+0.25$ across A1, A2, and B, despite large changes in accuracy, INCONCLUSIVE rate, and action distribution. This stability is consistent with findings that RLHF-aligned models develop systematic overconfidence as a byproduct of human preference optimization, where models learn to project higher certainty to satisfy rater preferences regardless of actual task difficulty (Kadavath et al., 2022), exposing a rigid decoupling between internal confidence and external behavioral control. Crucially, the declared error probabilities are identical to three decimal places between A2 and B (GPT: 0.146, Llama: 0.205). This provides direct evidence that the threshold table manipula-

Protocol	Model	Decl.	Actual	Gap
A0 (orig.)	GPT	n/r	n/r	n/r
	Llama	n/r	n/r	n/r
A1 (tokens)	GPT	0.235	0.488	+0.253
	Llama	0.301	0.369	+0.068
A2 (evid.)	GPT	0.146	0.410	+0.264
	Llama	0.205	0.433	+0.228
B (no thr.)	GPT	0.146	0.400	+0.254
	Llama	0.205	0.423	+0.218

Table 4: Calibration gap (actual error rate − declared probability) across protocols. A positive gap indicates overconfidence. A0 is marked n/r (not reported): declared error probabilities are not reliably captured under `max_tokens=80` truncation. GPT's gap is stable at $\approx +0.25$ across A1, A2, and B; Llama's gap increases substantially when full evidence is provided.

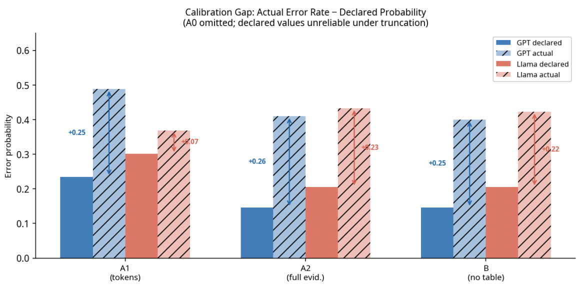

Figure 2: Calibration gap: actual error rate minus declared probability (positive = overconfident). GPT's gap is stable at $\approx +0.25$ across A1, A2, and B; Llama's gap is less stable. A0 is omitted (declared probabilities not reliably captured due to truncation).

tion is perfectly isolated to the control stage (Task 3) and does not leak backward to alter the monitoring stage (Task 2).

4.5 Finding 5: Models Systematically Under-Detect Genuine Ambiguity

In the stratified sample (gold INCONCLUSIVE rate = 33.3%), both models substantially under-predict the INCONCLUSIVE class (Table 5). Both models miss 75–88% of cases that expert annotators judged to be genuinely inconclusive. This is the opposite of the over-hedging narrative suggested by Protocol A0 (where INCONCLUSIVE rates of 48–79% appeared): that apparent over-hedging was an artifact of evidence truncation causing parse defaults. With full evidence and correct parsing, both models are under-hedging—committing to definitive answers on evidence that experts found ambiguous. In a clinical decision-support context, this is a safety-relevant finding.

Model	Pred. INC%	Recall	Precision
GPT (A2)	15.0%	25.0%	55.6%
Llama (A2)	10.7%	12.0%	37.5%

Table 5: INCONCLUSIVE class statistics (Protocol A2, stratified sample). Gold rate = 33.3%. Both models miss the majority of genuinely ambiguous cases.

5 A Reporting Checklist for Metacognitive Evaluation Pipelines

Based on the five findings above, we propose the following six fields that should be reported in any evaluation of LLM metacognitive control:

1. **Parse success rate per model.** Report the fraction of responses successfully parsed into structured output. Aggregate accuracy over all responses (including parse defaults) without disclosure is misleading. (As seen with Gemini-2.5-Flash, 99.8% parse failure can masquerade as 0.1% accuracy).
2. **`max_tokens` and its adequacy.** Report the token budget and provide evidence that it is sufficient for the expected response format. A smoke test (5–10 responses inspected manually) should be standard.
3. **Evidence length and truncation policy.** Report the maximum evidence length and the fraction of cases where evidence was truncated.
4. **Whether the action rule is supplied in the prompt.** If the prompt encodes a mapping from monitoring output to control action, report this explicitly and note that control-stage metrics are algebraic consequences, not behavioral measurements.
5. **Baselines.** Report majority-class and random baselines. For stratified samples, both equal $1/K$ where K is the number of classes. For natural samples, the majority class baseline must be calculated from the empirical distribution.
6. **Cohen's κ alongside raw agreement.** Raw agreement is inflated when one action dominates. Report κ for all agreement statistics.

6 Discussion

The threshold table as a homogenizer. The most counterintuitive finding is that the threshold table does not merely create a tautology—it also homogenizes models. With the table, both GPT and Llama achieve $\kappa = 1.000$ in A2 (GPT shows a single deviation in A1, $\kappa = 0.998$). Without it, they diverge to $\kappa = 0.860$ and $\kappa = 0.406$ respectively. A pipeline that supplies the action rule is not measuring two different models' control policies—it is measuring whether both models can follow the same instruction, which they can.

A general principle. When a prompt supplies an explicit mapping between two measured variables, one variable collapses into a deterministic function of the other and ceases to be an independent measurement. This principle applies beyond the specific pipeline studied here: any evaluation that provides the model with a scoring rubric, a decision rule, or a threshold table during inference should verify that the downstream metric still captures variance attributable to the model rather than to the supplied rule.

Monitoring is stable; control is malleable. The calibration gap ($\approx +0.25$ for GPT across all variants) suggests that the monitoring stage reflects a relatively stable model property. The control stage, by contrast, shifts by 28.7 percentage points in response to a preprocessing change (evidence length and sample stratification combined). Researchers who report only control-stage metrics without isolating the monitoring stage may be measuring prompt sensitivity rather than model capability.

7 Limitations

This study has several limitations that should be noted.

Scope and Generalizability. All empirical findings are based on two models (GPT-4.1-mini and Llama-3.3-70b) evaluated on the biomedical domain via the PubMedQA dataset. While PubMedQA serves as a rigorous diagnostic stress-test to demonstrate the existence of pipeline-manufactured artifacts, evaluating these dynamics across diverse benchmarks, non-medical domains, and smaller-scale open-weight models remains essential for future work.

Prompt Structural Design. Our protocols share a common core three-task prompt family. We did not evaluate alternative prompt paradigms, such as few-shot in-context learning, chain-of-thought reasoning, or different structural configurations of the input layout. In particular, all instances embed questions strictly within a designated `CLAIM` slot; isolating the behavioral variance introduced by question-versus-claim framing constitutes an important direction for future investigation and helps prevent superficial evaluation conclusions.

Absence of Human Baseline. We do not establish human performance thresholds or expert baseline detection rates for the `INCONCLUSIVE` class within our stratified sample. Consequently, while we show that models systematically under-detect genuine clinical ambiguity compared to gold labels, we cannot precisely quantify how far below expert human clinical judgment these models fall.

8 Conclusion

We have shown that five findings—a 38-point accuracy artifact, an abstention artifact driven 88% by evidence length and sample composition, complete elimination of control-stage variance via threshold tables, a stable calibration gap, and systematic under-detection of ambiguity—all emerged from a single evaluation pipeline applied to two models on one dataset. The most consequential finding for the field is that supplying an explicit action-mapping table in the prompt renders control-stage metrics algebraically redundant. We propose a six-field reporting checklist to prevent these confounds from propagating into future work.

The broader lesson is not that metacognitive evaluation is impossible, but that it requires the same care applied to any measurement instrument: explicit isolation of variables, reporting of parse statistics, and verification that the metric captures model variance rather than prompt variance.

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